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A computer simulation study of the dielectric properties of a model of methyl cyanide

I. The rigid dipole case

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The dielectric permittivity of a model fluid representing methyl cyanide has been studied with a view to testing a recently developed theory of the frequency dependent dielectric permittivity. Although there are large correlations between the orientations of dipoles the Kirkwood g factor is close to the trivial value of unity; this contrasts with the behaviour of Stockmayer and similar model fluids and is a consequence of the molecular shape. The time dependence of both the longitudinal and transverse components of the dipole density is obtained and related to the relaxation of the single molecule dipole correlation function. The applicability of the 'dipolaron' description to the oscillation observed in the relaxation of the longitudinal component is examined.

1. INTRODUCTION

Recently, a molecular theory [1] of the frequency dependent dielectric permittivity has been developed which avoids the introduction of cavities and other concepts of continuum electrostatics. The theory is cast in the same terms as are used to interpret other dynamical studies of liquids, such as light scattering, yet at the same time makes contact with results of classical dielectric theory by re-deriving well established results on local field factors, dielectric friction etc. within the molecular representation. The central problems for a dielectric theory of a polar fluid are the necessity of describing the effect of correlation between molecular dipole moments and of including the interaction-induced dipoles which are present in any condensed phase. The objective of this paper and the following one [2] is to examine the applicability of the theory to the dielectric properties of a computer simulated fluid. The attraction of the computer simulation for this purpose is that the two fundamental problems alluded to above can be separated. The present paper is concerned with the idealized

fluid of rigid dipoles where the interest is in the description of the reorientational dynamics in the presence of orientational correlation between molecules. In the succeeding paper we examine the effect on the dielectric properties of the additional induced dipoles and the success of attempts to describe their influence through the introduction of local field factors. To date most simulations of dielectric properties have studied simple potential models, for reason of speed. We are aware of no other study of the effect of induced dipoles in a polar fluid.

The model fluid was selected to resemble methyl cyanide (MeCN) at its normal liquid density as this is a highly polar system in which the density, reorientation times etc. may be taken as representative of a widely studied class of real fluids. A potential which reproduces experimental diffraction, dynamical and thermodynamic data on liquid methyl cyanide has been developed in earlier studies [3]; this work has shown that there is a high degree of local orientational order in the fluid.

To simulate dielectric properties successfully, care must be taken to deal consistently with the long-range coulombic intermolecular interactions both in solving the equations of motion and in relating the calculated quantities to the permittivity. The underlying issues have been carefully described in recent reviews [4, 5] and we shall summarize the results on how to relate the permittivity to dipole correlation functions in § 2. If the calculated dielectric properties are to resemble those of a real fluid it is necessary to include a large number of molecules, so that the local fluid structure becomes insensitive to the periodic boundary conditions, and to run the simulation for a long time so that the averages of the collective correlation functions are properly established [6]. So as to satisfy these criteria we have developed a simple three site model of MeCN which enabled us to include 500 molecules in the simulation and run for approximately 40 dielectric relaxation times. The model was developed from the more realistic six-site model studied earlier [3] in such a way as to preserve the local structure of the fluid. We shall describe this potential and briefly compare calculated fluid properties for the three- and six-site models in § 3.

In § 4 we consider the static permittivity. We shall be particularly concerned to show that the calculated permittivity is behaving as if it were a property of a normal fluid and that different routes to the permittivity lead to consistent results. A very extensive study of this kind for the Stockmayer fluid has been made [7]. Section 5 deals with the relaxation of the permittivity and with the applicability of theory [1] to the effect of dipole-dipole correlation on the relaxation times. We shall compare carefully the relaxation of the transverse and longitudinal components of the dipole density; the latter shows a prominent oscillatory feature and we shall discuss whether this should be attributed to a librational oscillation of the molecules or to a collective 'dipolaron' oscillation, as predicted by Lobo *et al.* [8] and observed in simulations of Stockmayer fluids by Alder and Pollock [9].

We shall postpone a comparison between the dielectric properties of the simulated and real fluids until the next paper, where we deal with the more realistic model which includes the induced dipole effects.

2. COMPUTER SIMULATION AND THE DIELECTRIC PERMITTIVITY

The relationship between properties calculable in a simulation and the dielectric properties of real fluids has been discussed in reviews by Alder and

Pollock [5] and Stell *et al.* [4]. A computer simulation with cubic periodic boundary conditions and an Ewald treatment of the coulombic interactions mimics a physical system consisting of an arbitrarily large sphere of periodically repeated simulation cubes embedded in a continuum of infinite dielectric constant [10]. The dielectric properties of this periodic system may be related, via linear response theory, to the fluctuations in appropriately constructed dipole densities, which may be calculated in the simulation. For finite wavelength fluctuations, the effect of the surrounding dielectric is immaterial [7]. However, these dielectric properties may only be related to those of a real fluid if the computer simulation contains enough particles for the range (ξ) of the intermolecular correlations which determine the permittivity to be significantly smaller than the repeat distance (L) of the periodic system. The effect of periodic boundary conditions on the orientational correlation between non-polar molecules has been demonstrated [11]. We envisage a minimum condition on L for the periodicity to play no role in the calculated dielectric properties of

$$2\xi/L < 1. \quad (2.1)$$

Since the permittivity is an intensive, material property, independent of sample shape and size it must be true that its value is determined by intermolecular correlations which are short compared with any macroscopic length; that is, ξ must be of the order of several molecular diameters.

For an infinite system the permittivity tensor, ϵ , is given by

$$\mathbf{P}(\mathbf{k}, \omega) = \epsilon_0(\epsilon(\mathbf{k}, \omega) - 1) \cdot \mathbf{E}(\mathbf{k}, \omega), \quad (2.2)$$

where $\mathbf{P}(\mathbf{k}, \omega)$ and $\mathbf{E}(\mathbf{k}, \omega)$ are the spatial and temporal fourier components of the polarization and macroscopic ('Maxwell') field. For the periodic cubic system the values of $\mathbf{k}$ are restricted to

$$\mathbf{k} = (2\pi/L)(m, n, p), \quad (2.3)$$

where m, n, p are integers. In a system of cubic or spherical symmetry ϵ must be of the form

$$\epsilon(\mathbf{k}, \omega) = \epsilon_T(k, \omega)(1 - \hat{\mathbf{k}}\hat{\mathbf{k}}) + \epsilon_L(k, \omega)\hat{\mathbf{k}}\hat{\mathbf{k}} \quad (2.4)$$

since this is the only symmetric, second rank tensor which may be formed from $\mathbf{k}$. Because of the short-ranged nature of the correlations which determine ϵ , $\epsilon(\mathbf{k}, \omega)$ must become $\mathbf{k}$ independent when k is sufficiently small [1], that is

$$\epsilon(\mathbf{k}, \omega) = \epsilon(\omega)\mathbf{I}, \quad k\xi < 1. \quad (2.5)$$

For the systems of cubic or spherical symmetry of interest here, equation (2.5) implies that

$$\epsilon_T(k, \omega) = \epsilon_L(k, \omega) = \epsilon(\omega), \quad k\xi < 1. \quad (2.6)$$

The application of linear response theory to the Maxwell field is not straightforward [12, 1]; it is preferable to obtain molecular expressions for ϵ indirectly by calculating the susceptibility (χ^0) which gives the polarization as a response to the external field ($\mathbf{E}^0$), i.e.

$$\mathbf{P}(\mathbf{k}, \omega) = \epsilon_0\chi^0(\mathbf{k}, \omega) \cdot \mathbf{E}^0(\mathbf{k}, \omega). \quad (2.7)$$

In an infinite system the external and Maxwell fields are related by [1]

$$\mathbf{E}(\mathbf{k}, \omega) = \mathbf{E}^0(\mathbf{k}, \omega) - \epsilon_0^{-1} \hat{\mathbf{k}} \hat{\mathbf{k}} \cdot \mathbf{P}(\mathbf{k}, \omega) \quad (2.8)$$

so that if we combine equations (2.4), (2.7) and (2.8), in a frame in which $\hat{\mathbf{k}}$ defines the z direction, we obtain

$$\epsilon_T(k, \omega) - 1 = \chi_{xx}^0(\mathbf{k}, \omega), \quad (2.9 a)$$

$$(\epsilon_L(k, \omega) - 1) \epsilon_L(k, \omega)^{-1} = \chi_{zz}^0(\mathbf{k}, \omega). \quad (2.9 b)$$

Linear response theory relates χ^0 to correlation functions of the Fourier components of the dipole density of the system ($\mathbf{M}(\mathbf{k}, t)$). $\mathbf{M}$ is conveniently resolved into its longitudinal and transverse components

$$\mathbf{M}(\mathbf{k}, t) = \mathbf{M}_L(\mathbf{k}, t) + \mathbf{M}_T(\mathbf{k}, t), \quad (2.10)$$

where

$$\mathbf{M}_L(\mathbf{k}, t) = \sum_i^N \hat{\mathbf{k}} \hat{\mathbf{k}} \cdot \boldsymbol{\mu}^i(t) \exp(i\mathbf{k} \cdot \mathbf{r}^i(t)) \quad (2.11 a)$$

and

$$\mathbf{M}_T(\mathbf{k}, t) = \sum_i^N (1 - \hat{\mathbf{k}} \hat{\mathbf{k}}) \cdot \boldsymbol{\mu}^i(t) \exp(i\mathbf{k} \cdot \mathbf{r}^i(t)) \quad (2.11 b)$$

and the sums run over all molecules i with dipole moment $\boldsymbol{\mu}^i$ and position $\mathbf{r}^i$. The necessary components of χ^0 are given by ($\mathbf{k}$ parallel to z)

$$\begin{aligned} \chi_{xx}^0(\mathbf{k}, \omega) = (1/2) \lim_{N, V \rightarrow \infty} (k_B T V \epsilon_0)^{-1} \langle \mathbf{M}_T(\mathbf{k}, 0) \cdot \mathbf{M}_T(-\mathbf{k}, 0) \rangle \\ \times \left\{ 1 + i\omega \int_0^\infty dt \exp[i\omega t] \Phi_T(\mathbf{k}, t) \right\}, \end{aligned} \quad (2.12 a)$$

and

$$\begin{aligned} \chi_{zz}^0(\mathbf{k}, \omega) = \lim_{N, V \rightarrow \infty} (k_B T V \epsilon_0)^{-1} \langle \mathbf{M}_L(\mathbf{k}, 0) \cdot \mathbf{M}_L(-\mathbf{k}, 0) \rangle \\ \times \left\{ 1 + i\omega \int_0^\infty dt \exp[i\omega t] \Phi_L(\mathbf{k}, t) \right\}, \end{aligned} \quad (2.12 b)$$

where

$$\Phi_A(\mathbf{k}, t) = \langle \mathbf{M}_A(\mathbf{k}, t) \cdot \mathbf{M}_A(-\mathbf{k}, 0) \rangle / \langle |\mathbf{M}_A(\mathbf{k}, 0)|^2 \rangle. \quad (2.12 c)$$

Equations (2.6), (2.9) and (2.12) give a possible route to $\epsilon(\omega)$, since the correlation functions of $\mathbf{M}_L$ and $\mathbf{M}_T$ may be obtained in the simulation. However, the values of $\mathbf{k}$ which may be studied are restricted by the periodicity of the system (cf. equation (2.3)) so that the smallest wavevector which may be studied ($k_{\min}$) is of length $2\pi/L$. In order that the condition $k\xi < 1$ be satisfied it is necessary to use a large system. Whether this low k limit is reached can be tested through the consistency of the calculated values of ϵ_L and ϵ_T with equation (2.6). We note that if the (testable) condition $k_{\min}\xi < 1$ is satisfied then, by equation (2.1), the system will be large enough for us to identify the calculated $\epsilon(\omega)$ with a property of an isotropic fluid of the same molecules.

Most computer simulation studies of dielectric properties have approached the calculation of the permittivity through the calculation of the total moment of the primitive cube of the simulation, i.e.

$$\mathbf{M} = \sum_i \boldsymbol{\mu}^i \quad (2.13)$$

where i runs over the molecules in the primitive cube. For a simulation in which the long-ranged part of the coulombic interactions has been dealt with by the Ewald method the relationship between the permittivity and the correlation function of this moment is [4, 5]

$$\epsilon^E(\omega) - 1 = (3k_B T \epsilon_0 L^3)^{-1} \langle |\mathbf{M}|^2 \rangle \left\{ 1 + i\omega \int_0^\infty dt \exp[i\omega t] \Phi_{\mathbf{M}}(t) \right\} \quad (2.14)$$

ϵ^E is the $k \equiv 0$ permittivity of the periodic system studied in the computer simulation. It may be identified with the permittivity of an isotropic fluid only if equation (2.1) is satisfied.

3. THE COMPUTER SIMULATION MODEL

The purpose of this study was to investigate the dielectric properties of a realistic model system in which the intermolecular potential and the intermolecular distribution function of the fluid are strongly affected by molecular shape. However, as described in the last section, in order to extract a meaningful permittivity from the calculations it is necessary to study a large system and to treat correctly the long-range coulombic interactions; these features limit the sophistication of the potential model in order to keep the computation time within reasonable bounds.

Elsewhere [3] a six-site model of MeCN has been described which gives an excellent description of the structural and dynamical properties of the fluid. In that model the interaction sites are at the atomic positions, they consist of suitably parameterized Lennard-Jones 12-6 functions and point charge interactions. The point charges were chosen to reproduce the electrostatic potential obtained in an *ab initio* calculation [13]. For our present purposes we have simplified this model to a 3-site one, by replacing the methyl group carbon and hydrogen interaction sites by a single site (Me) located on the methyl carbon atom, so that

$$u_{ij} = \sum_{\alpha=1}^3 \sum_{\beta=1}^3 \left[4\epsilon_{\alpha\beta} \left\{ \left(\frac{\sigma_{\alpha\beta}}{r_{i\alpha,j\beta}} \right)^{12} - \left(\frac{\sigma_{\alpha\beta}}{r_{i\alpha,j\beta}} \right)^6 \right\} + \frac{q_\alpha q_\beta}{r_{i\alpha,j\beta}} \right], \quad (3.1)$$

where α and β label sites on molecules i and j . The carbon-carbon and nitrogen-nitrogen Lennard-Jones parameters have been kept at their values in the six-site model. The Me-Me parameters were adapted from a 20-6 potential which describes methane [14]. The Lennard-Jones parameters for the cross interactions were taken from the usual mixing rules [14]. The point charges on all the interaction sites were recalculated so as to keep the molecular dipole moment and quadrupole moment the same as in the six-site model (i.e. 4.14 D and -0.53 eÅ^2 at the centre-of-mass, respectively). The parameter values are given in table 1. Certain properties of the three- and six-site models are compared below.

Table 1. Parameters of the 3-site potential.

	$\epsilon_{\alpha\beta}/\text{K}$	$\sigma_{\alpha\beta}/\text{\AA}$	q_{α}/e
N	50	3.3	-0.398
C	50	3.4	0.129
Me	191	3.6	0.269

$$r_{\text{MeC}} = 1.46 \text{ \AA}, \quad r_{\text{MeN}} = 2.63 \text{ \AA}.$$

With this simplified model it proved possible to simulate 500 molecules for a time of 100 ps. The rotational equations of motion were solved with the Singer-Taylor method [15] and integrated using the Verlet leapfrog algorithm; the program was adapted from the 3-site diatomic program of McDonald *et al.* [16]. (The details of the modification to 3 mass points are given elsewhere [17].) Two simulation runs were made at a molar volume of 52.18 cm^3 (the density of MeCN at atmospheric pressure and a temperature of 293 K). A large system of 500 molecules was studied on the Cray 1 computer at the Daresbury Laboratory. Starting from a lattice, the system was equilibrated for 37 ps and the correlation functions required for the dielectric studies were calculated over the remaining 63 ps of the run. The mean temperature was 307 K. A timestep of 10^{-14} s was used, which gave an energy drift of less than 0.1 per cent over the whole run. Because of the difficulty of accessing magnetic tapes on this computer the correlation functions were calculated in parallel with the simulation run itself. So as to keep the restart file at a manageable size only the collective correlation functions (i.e. of $\mathbf{M}_L$, $\mathbf{M}_T$ and $\mathbf{M}$) were obtained in this calculation. In order to obtain single particle properties a smaller run with 108 molecules was run on the Cambridge University IBM 3081 for a period of 68 ps, configurations were dumped to tape and subsequently analysed by conventional methods. The mean temperature of this run was 312 K. In both simulations the electrostatic contributions to the energy and the forces were evaluated by the Ewald method. In the larger run, from which the dielectric properties were obtained, the series was truncated at 754 wavevectors (so that the largest wavevector included is at $1.24 \text{ \AA}^{-1}$). In both cases the cut-off distance for the short-range interactions was one half of the box length; because of the different system sizes this means that the potentials used in the two calculations are not quite the same, but the differences are unlikely to affect the values of those properties of the runs which we wish to compare.

In figure 1 we show a comparison of the molecular structure of the fluids given by the six- and three-site models. The functions plotted, denoted G_0 , G_1 and G_2 , relate respectively to the packing of the centres of mass of neighbouring molecules and to the local dipole and quadrupole orientational order of the fluid. To be specific, G_0 is just the normal radial distribution function and G_1 and G_2 are given by

$$G_1(r) = \overline{P_1(\cos \theta_{12})}[r], \quad (3.2 a)$$

and

$$G_2(r) = 2\overline{P_2(\cos \theta_{12})}[r], \quad (3.2 b)$$

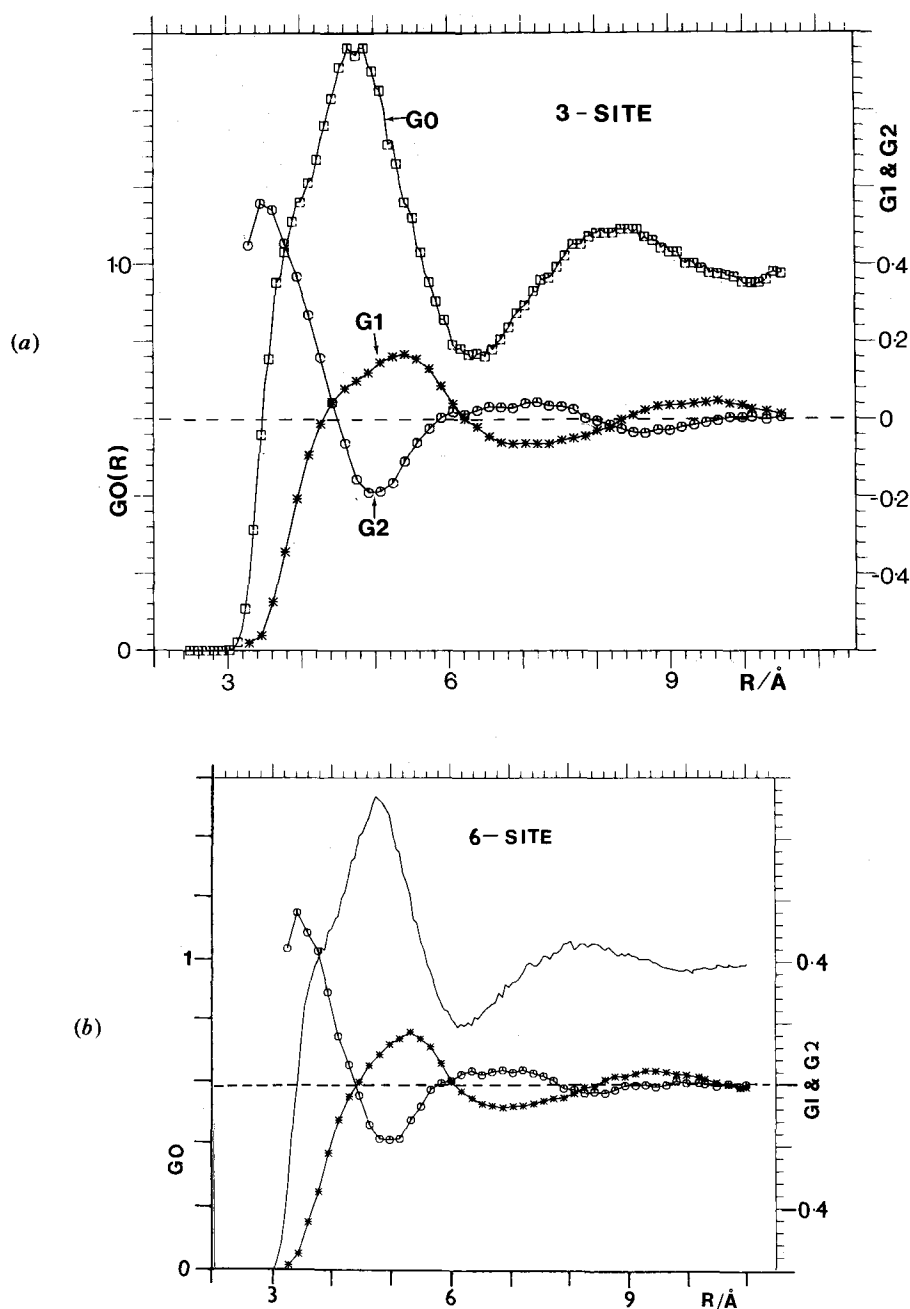


Figure 1. Centre of mass distribution functions G_0 , G_1 and G_2 {cf. equation (3.2)} for the 3-site model of the present work compared to the 6-site model of [3 (a)].

where P_n is a Legendre polynomial of rank n , θ_{12} is angle between the symmetry axes of molecules 1 and 2 which are separated by the distance r and the overbar denotes an average. The agreement between the structural properties of the three- and six-site model is seen to be excellent (we recall that the latter model

gives very good agreement with the diffraction data on liquid MeCN [18]). The relationship of these functions to a model of the molecular packing of the fluid has been discussed at length in [3 (a)].

The ‘invariant’ expansion of the molecular pair distribution function, $g(\mathbf{r}_{12}, \Omega_1, \Omega_2)$ (where Ω_i denotes the Euler angles which specify the orientation of molecule i) in an isotropic fluid, is given by [19]

$$g(\mathbf{r}_{12}, \Omega_1, \Omega_2) = \sum_{\substack{l_1, l_2, l_{12} \\ m_1, m_2}} i^{l_1-l_2-l_{12}} g^{l_1 l_2 l_{12}}(\mathbf{r}_{12}) \begin{pmatrix} l_1 & l_2 & l_{12} \\ m_1 & m_2 & m_{12} \end{pmatrix} \times C_{l_1 m_1}(\Omega_1) C_{l_2 m_2}(\Omega_2) C_{l_{12} m_{12}}(\hat{\mathbf{r}}_{12}), \quad (3.3)$$

where $C_{l,m}$ denotes a spherical harmonic and we have specialized to the case of linear molecules. The dielectric properties of the fluid are determined by $g^{110}(r)$ [1]. In terms of the coefficients of the invariant expansion we have

$$G0(r) = g^{000}(r), \quad (3.4 a)$$

$$G1(r) = 3^{-3/2} g^{110}(r)/g^{000}(r), \quad (3.4 b)$$

$$G2(r) = 5^{-3/2} g^{220}(r)/g^{000}(r). \quad (3.4 c)$$

Our principal concern in devising the three-site model was to obtain a fluid structure in agreement with that of the six-site model ; the agreement between

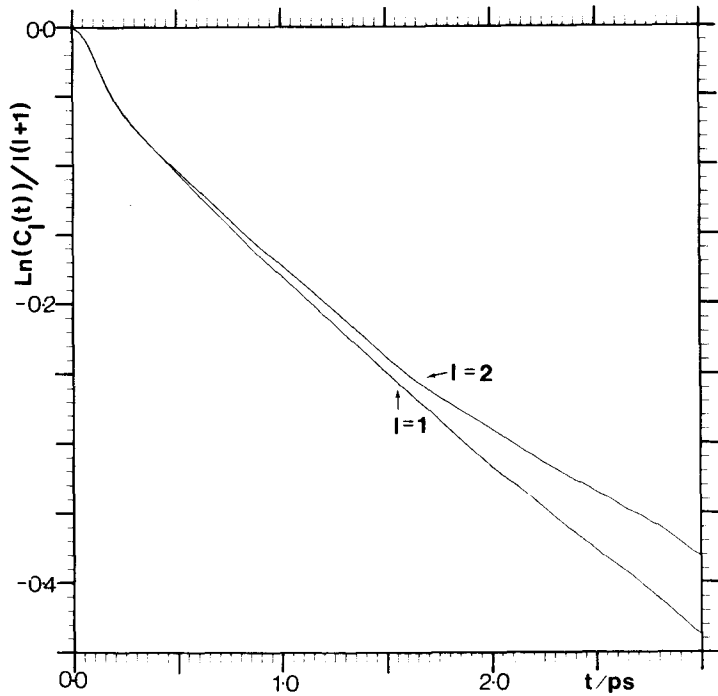


Figure 2. The single molecule $l=1$ and $l=2$ orientational correlation functions plotted in reduced form $\text{Ln } C_l/(l(l+1))$. The corresponding functions for the six-site model are discussed in [3 (b)].

the thermodynamic and dynamical properties of the two models is less satisfactory. The total potential energy was found to be $-34.2 \text{ kJ mol}^{-1}$ (for the Cray run) compared with $-28.7 \text{ kJ mol}^{-1}$ [3 (a)] for the six-site model; however, the coulombic contributions to the energy in the two calculations were in close agreement. The pressure of the 3-site model was rather high ($PV/RT = 1.5$ for the Cray run). The diffusion coefficient was found to be $3.2 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ at 307 K, compared with the value of $4.3 \times 10^{-5} \text{ cm}^2 \text{ s}^{-1}$ at 291 K for both the six-site model [3 (b)] and the real fluid [20]. The first and second rank, single molecule, orientational correlation functions ($C_i(t)$) for the three-site model are displayed in figure 2. The shape of these functions is very similar to that found in the six-site model of MeCN [3 (b)] and in other fluids in which the orientational motion is strongly hindered [21]. The correlation time of the first rank function (defined as the integral over the normalized correlation function) is found to be 3.7 ps at 312 K, compared with 2.8 ps at 291 K for the six-site model, which gives reorientation times in agreement with N.M.R. and Raman [3]. This discrepancy and that of the diffusion coefficients indicates that the molecular motion is more strongly hindered in the three-site model than in either the six-site model or the real fluid [22].

4. STATIC DIELECTRIC PROPERTIES

As discussed in § 2, the longitudinal and transverse components of the permittivity tensor are related to the correlation functions of the longitudinal and transverse dipole densities. For the static ($\omega = 0$) case, equations (2.12) reduce to

$$\epsilon_T(k) - 1 = y \left\{ \frac{\langle |\mathbf{M}_T(\mathbf{k})|^2 \rangle}{2N\mu^2} \right\} \quad (4.1 a)$$

and

$$\frac{\epsilon_L(k) - 1}{\epsilon_T(k)} = y \left\{ \frac{\langle |\mathbf{M}_L(\mathbf{k})|^2 \rangle}{N\mu^2} \right\}, \quad (4.1 b)$$

where

$$y = \rho\mu^2/k_B T\epsilon_0.$$

For the dipole appropriate to the simulation model

$$y = 58.0. \quad (4.2)$$

We have calculated the bracketed quantities of equations (4.1) at the three smallest k values accessible in the simulation: $k = k_{\min}$, $\sqrt{2}k_{\min}$ and $\sqrt{3}k_{\min}$, where $k_{\min} = 2\pi/L$ ($= 0.179 \text{ \AA}^{-1}$). For these values of k there are, respectively, 3, 6 and 4 wavevectors which are equivalent under the symmetry operations of the cube. By averaging the computed functions over wavevectors corresponding to a given k we may improve the statistics of the permittivity values. Furthermore, we may estimate the statistical uncertainty in these mean correlation functions by calculating the standard deviations over the same sets. The values of the mean correlation functions at each k value are shown in table 2; the standard deviations are shown in parentheses.

By substituting the values of the transverse correlation function into equation (4.1 a) we arrive at the values for ϵ_T given in the third column of table 2. For large values of the permittivity the longitudinal equation (4.1 b) is numerically

Table 2. The static permittivity.

$k/k_{\min}$	$\frac{\langle \mathbf{M}_T(\mathbf{k}) ^2 \rangle}{2N\mu^2}$	$\frac{\langle \mathbf{M}_L(\mathbf{k}) ^2 \rangle}{N\mu^2}$	ϵ_T	ϵ^R
1	0.45 (0.05)	0.018 (0.002)	27	26
2	0.46 (0.09)	0.018 (0.002)	28	25
3	0.54 (0.05)	0.020 (0.002)	33	33

unstable with respect to errors in the correlation function. Consequently we are unable to test directly the equality of ϵ_L and ϵ_T which is expected to occur at sufficiently small k (equation (2.6)). However if the condition $k\xi < 1$ were satisfied then we would have

$$\epsilon = \langle |\mathbf{M}_T(\mathbf{k})|^2 \rangle / (2 \langle |\mathbf{M}_L(\mathbf{k})|^2 \rangle). \quad (4.3)$$

Values of this ratio, denoted ϵ^R , are given in the fourth column of table 2. These results are consistent with the calculated values of ϵ_T , within the statistical error limits.

The alternative route to the static permittivity (at $k \equiv 0$) is via equation (2.14)

$$\epsilon^E - 1 = y [\langle |\mathbf{M}|^2 \rangle / 3N\mu^2]. \quad (4.4)$$

The simulation yields a value of 0.55 for the quantity in square brackets, which corresponds to a value for ϵ^E of 33. The likely errors in this quantity are larger than those associated with the calculations at finite k because the latter values are averages over several independent 'measurements'.

The results of our calculations are consistent with the assertion that the system size of the simulation is sufficiently large to satisfy the limits $k_{\min}\xi < 1$, $2\xi/L < 1$. This statement is supported by the agreement of the values of ϵ^R and ϵ^T , by the lack of a pronounced k dependence in the calculated values of $\epsilon_T(k)$ and by their good agreement with ϵ^E . In view of this we shall proceed by discussing the calculated correlation functions as if they represent the equivalent functions in an isotropic fluid.

In an infinite, isotropic fluid the longitudinal and transverse dipole density correlation functions may be related to integrals over the coefficients of the invariant expansion of the distribution function, i.e. [1]

$$\langle |\mathbf{M}_T(\mathbf{k})|^2 \rangle / 2N = (y/3) \{g^S(k) - g^L(k)\}, \quad (4.5 a)$$

$$\langle |\mathbf{M}_L(\mathbf{k})|^2 \rangle / N = (y/3) \{g^S(k) + 2g^L(k)\}, \quad (4.5 b)$$

where

$$g^S(k) = 1 - (4\pi\rho 3^{-3/2}) \int_0^\infty dr r^2 j_0(kr) g^{110}(r), \quad (4.6 a)$$

and

$$g^L(k) = 4\pi\rho(180)^{-1/2} \int_0^\infty dr r^2 j_2(kr) g^{112}(r), \quad (4.6 b)$$

where $j_n(x)$ is a spherical Bessel function of order n . $g^{110}(r)$ is a short-ranged function of range ξ , whereas g^{112} decays like r^{-3} at large r [1, 4, 5]. g^L may be eliminated from equations (4.5) with the result

$$N^{-1}[\langle |\mathbf{M}_T(\mathbf{k})|^2 \rangle + \langle |\mathbf{M}_L(\mathbf{k})|^2 \rangle] = y g^S(k). \quad (4.7)$$

In the low k limit (i.e. $k\xi < 1$) the integral in (4.6 a) becomes k independent and [1]

$$\lim_{k \rightarrow 0} g^S(k) = 1 + \sum_{j \in S_1} \langle \cos \theta_{ij} \rangle, \quad (4.8)$$

where the sum runs over those molecules which lie within a sphere S_1 of radius greater than ξ around molecule 1, this is the normal representation of the Kirkwood g -factor. In this low k limit, using equations (2.6), (2.9) and (2.12) equation (4.7) becomes the Kirkwood relationship

$$\frac{(\epsilon - 1)(2\epsilon + 1)}{\epsilon} = y \lim_{k \rightarrow 0} g^S(k). \quad (4.9)$$

Values for $g^S(k)$ at the three k values accessible in the simulation are shown in table 3. The Kirkwood factor at $k \equiv 0$ may be calculated by substituting the value of the $k \equiv 0$ permittivity (i.e. ϵ^E) into the left hand side of equation (4.9). The result obtained in this way is denoted $g^S(k \equiv 0)$ in table 3. (Note that this value is *not* $\langle \mathbf{M}^2 \rangle / 3N\mu^2$ [4, 5]).

Table 3. 'Kirkwood' g -parameters.

$k/k_{\min}$	$g^S(k)$
$\equiv 0$	1.1
1	0.91
2	0.93
3	1.1

The $g^S(k)$ values show only an insignificant k dependence over the range of k which we have studied and agree with the value of $g^S(k \equiv 0)$. This behaviour is consistent with $k\xi < 1$ and our assumption that the system size we have used is sufficiently large to study dielectric properties.

It is clear from the behaviour of $G1$, in figure 1, that there *are* important dipole-dipole correlations in the simulated fluid. However the values of $g^S(k)$ calculated above lie close to unity. This behaviour contrasts markedly with that found in simulations of Stockmayer fluids and in other fluids in which the only orientation dependent contributions to the potential are dipole-dipole interactions [5, 23]. In the Stockmayer fluid g^{110} is positive almost everywhere so that large values of g^S are found. In the simulation of MeCN the large initial negative value of $G1$ and the large positive value of $G2$ shows that near neighbouring molecules strongly prefer an antiparallel arrangement, this is a consequence of the molecular shape [3 (a)]. The result of the subsequent oscillatory character of g^{110} is that the intermolecular correlation contributions to the permittivity are strongly quenched and the permittivity is much smaller than

would be found for a Stockmayer fluid with the same reduced dipole moment. The permittivity of the Stockmayer fluid increases faster than linearly with y (see figure 17 in [4]) and Pollock and Alder [5] report a value of $\epsilon = 80$ for a y value of 29.7. From these considerations we might expect ϵ to be about 200 for such a polar fluid as MeCN. Patey has shown that quadrupolar interactions lead to a large reduction in the value of ϵ from the Stockmayer predictions (figure 22 of [4]) and it has been suggested that such interactions are responsible for the success of the Onsager equation (i.e. $g^s = 1$) for real fluids. If we take as an effective diameter for MeCN $d = 4.7$ (as suggested by the shape of G_0 in figure 1), we find that for our simulation $\rho^* = 1.2$, $(\mu^*)^2 = 3.9$, and $(Q^*)^2 = 0.07$, where Q is the quadrupole moment of the model with respect to the centre of mass and $*$ denotes a reduced quantity. This quadrupole is too small to have a significant effect on the fluid structure, which reinforces our view that the dominant cause of the failure of the Stockmayer predictions is the influence of the molecular shape [23].

As we have stressed in earlier sections, we do not expect the calculated permittivity to be in good agreement with experiment ($\epsilon = 36$ [24]) because the three-site model has obvious deficiencies and because the present calculations do not include the interaction-induced dipoles which are present for any real, polarizable fluid. The latter defect is partially remedied in the following paper, where a more detailed comparison with experiment is given. The most we can do at present is to compare our value for g^s with the Kirkwood g -factors obtained from experiment. These values are subject to some assumption about the local field effects. Buckingham [24] has quoted a g -factor of 0.82, obtained by application of the Kirkwood–Fröhlich formula [1]. This may be compared with our best low k result ($k^2 = 1$) of $g^s = 0.91$, which is subject to less statistical uncertainty than the $k \equiv 0$ value but, nonetheless is only likely to be reliable to ~ 10 per cent.

5. TIME DEPENDENT CORRELATION FUNCTIONS

5.1. General considerations

The normalized, time dependent correlation functions $\Phi_T(k, t)$ and $\Phi_L(k, t)$ (cf. equation (2.12 c)) at the k values $k_{\min}$, $\sqrt{2}k_{\min}$ and $\sqrt{3}k_{\min}$ are shown in figures 3 and 4. The k dependence of these functions over this range of k is not great. The transverse function Φ_T is very similar in shape to the single molecule first rank orientational correlation function shown in figure 2. It decays almost exponentially with a decay constant of order 4 ps. The longitudinal function Φ_L shows a very rapid initial decay, followed by a pronounced oscillatory feature. The relationship between the Φ_L and Φ_T functions is similar to that which was found in a computer simulation of water [21 (b)], although the oscillatory character of the functions was more pronounced in that case.

The normalized correlation function Φ_M of the total moment of the sample $\{\mathbf{M}$, equation (2.13) $\}$ is also shown in figure 3 (offset along the time axis for clarity). For the computer simulation with Ewald summation of the coulombic interactions this correlation function is related to ϵ^E by the same functional as $\Phi_T(k, t)$ is related to $\epsilon_T(k, \omega)$ (cf. equations (2.12) and (2.14)). The similarity of the shapes of Φ_M and Φ_T thus shows that the frequency dependence of ϵ^E and of ϵ_T will agree, as well as their $\omega = 0$ values, discussed in § 4.

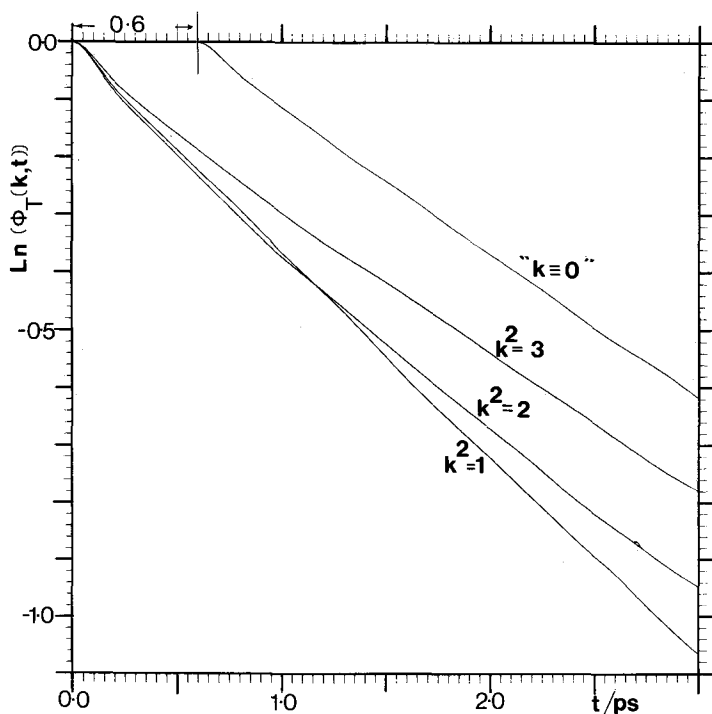


Figure 3. The collective transverse dipole density correlation functions (Φ_T) for $(k/k_{\min})^2 = 1, 2$ and 3 . Also shown (denoted $k \equiv 0$) is the normalized correlation function of the total dipole of the primitive cube of the simulation {cf. equation (2.14)}; this function is offset by 0.6 ps along the time axis, for clarity.

The real parts of the time transforms, $\Phi_L(k, \omega)$ and $\Phi_T(k, \omega)$, are compared in figure 5. These functions were obtained by the application of a Maximum Entropy procedure, as described in [21 (a)]. The oscillatory character of $\Phi_L(k, t)$ shows up as a very broad peak at ~ 100 cm^{-1} . The spectral density of Φ_T is comparatively featureless; the initial, non-exponential character of the time correlation function is responsible for a weak shoulder at approximately 80 cm^{-1} . In this section we shall apply current theories of dielectric relaxation to the interpretation of these data; Madden and Kivelson [1] have shown how a wide range of relaxation phenomena may be interpreted through a straightforward application of projection operator methods. These theories are appropriate to systems of isotropic or cubic symmetry but in the low k limit, we must therefore proceed with care in applying them to the computer simulation results. By applying Mori's generalized Langevin equation [25] to the variables $\mathbf{M}_L(\mathbf{k}, t)$ and $\mathbf{M}_T(\mathbf{k}, t)$ the following expressions for $\Phi_L(k, \omega)$ and $\Phi_T(k, \omega)$ are obtained [1]:

$$\Phi_L(k, \omega) = [-i\omega + R_L(k, \omega) / \langle |\mathbf{M}_L(\mathbf{k})|^2 \rangle]^{-1}, \quad (5.1 a)$$

$$\Phi_T(k, \omega) = [-i\omega + R_T(k, \omega) / \langle |\mathbf{M}_T(\mathbf{k})|^2 \rangle]^{-1}, \quad (5.1 b)$$

where

$$R_A(k, \omega) = \int_0^\infty dt \exp[i\omega t] \langle \dot{\mathbf{M}}_A(\mathbf{k}, t_{\mathbf{m}_A}) \cdot \dot{\mathbf{M}}_A(-\mathbf{k}, 0) \rangle, \quad (5.2)$$

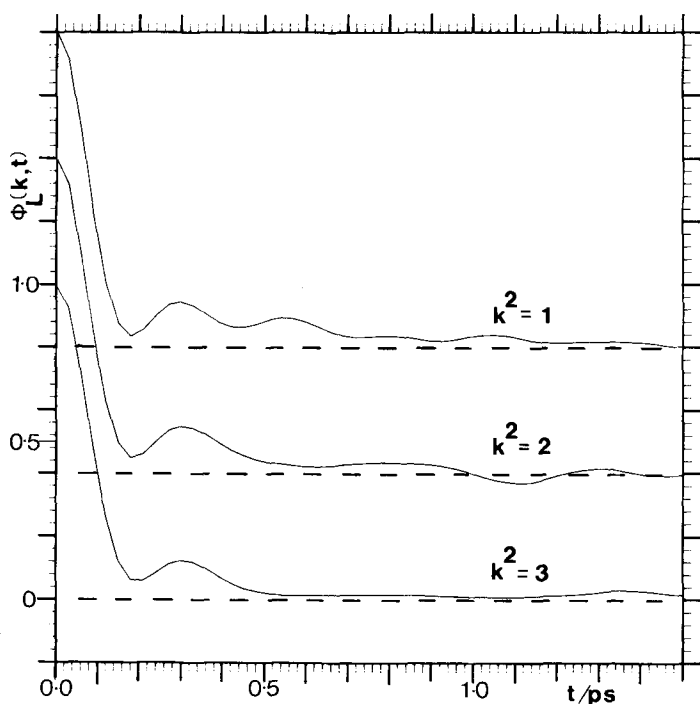


Figure 4. The collective longitudinal dipole density correlation functions for $(k/k_{\min})^2 = 1, 2$ and 3. The functions are offset along the vertical by 0.4 scale units, for clarity.

and $t_{\mathbf{M}_A}$ indicates that the time evolution is subject to the usual Mori projected Liouville operator [25]. The great advantage of this transformation is that it can be shown [1], when k is sufficiently small for $\epsilon_L(k, \omega)$ and $\epsilon_T(k, \omega)$ to be equal, that

$$R_L(k, \omega) = R_T(k, \omega)/2 \equiv R(\omega) \quad (5.3)$$

(the factor of 2 arises because the transverse dipole density contains two independent components). This shows that the projected time correlation function in equation (5.2) depends only on short range intermolecular correlations. This feature makes R a useful quantity with which to interpret dynamical data, by analogy with the use of the short-range Kirkwood factor {see (4.8)} to interpret the static permittivity.

In order to make contact with the permittivity we may make use of equations (4.1) and (2.6) to rewrite (5.1) as

$$\lim_{k \rightarrow 0} \Phi_L(k, \omega) = [-i\omega + (\tau_D(\omega))^{-1} \epsilon(0)]^{-1}, \quad (5.4 a)$$

$$\lim_{k \rightarrow 0} \Phi_T(k, \omega) = [-i\omega + (\tau_D(\omega))^{-1}]^{-1}, \quad (5.4 b)$$

where

$$(\tau_D(\omega))^{-1} = \{R(\omega)y/N\mu^2\}(\epsilon(0) - 1)^{-1}. \quad (5.5)$$

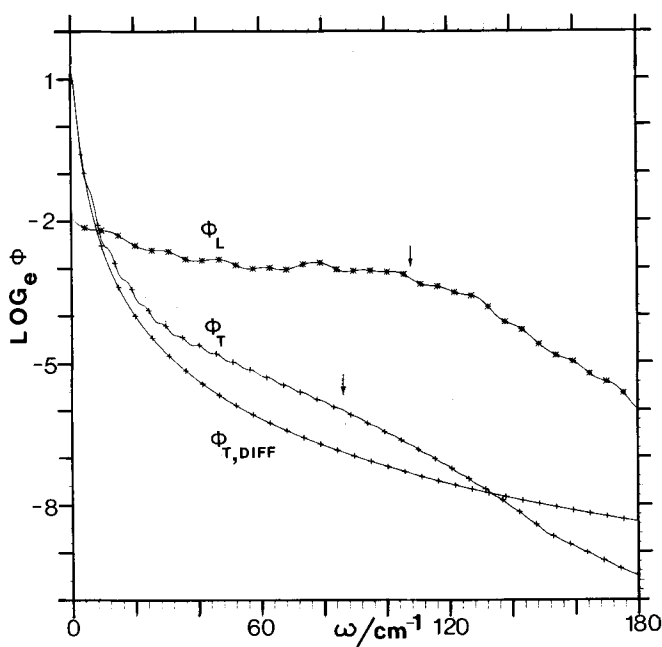


Figure 5. The spectral densities of the longitudinal (Φ_L) and transverse (Φ_T) dipole density correlation functions. Also shown is the prediction of the diffusion model ($\Phi_{T,DIFF}$) for the transverse function (cf. (5.4), with $\tau_D(\omega) = \tau_D(0)$). The arrows mark the positions of librational peaks in Φ_L and Φ_T , predicted by the Mori theory (see figure 6).

τ_D may be recognized as the Debye relaxation time, as substitution of (5.4 *b*) into (2.12 *a*) and (2.9 *a*) leads to

$$\frac{\epsilon(\omega) - 1}{\epsilon(0) - 1} = [1 - i\omega\tau_D(\omega)]^{-1}. \quad (5.6)$$

5.2. Low frequency regime

As a first application of these ideas we may relate the vastly different decay times of $\Phi_L(k, t)$ and $\Phi_T(k, t)$ (cf. figures 3 and 4). Suitable decay times may be defined as the time integrals of the functions, they correspond to $\Phi_L(k, \omega = 0)$ and $\Phi_T(k, \omega = 0)$. Values of these times were obtained by numerically integrating the correlation functions to 5 ps and adding the integral from 5 ps to infinity of an exponential which fits the function between 3 and 5 ps, the values are given in table 4. According to equations (5.1)

$$\frac{\Phi_A(\mathbf{k}, \omega = 0)}{\langle |\mathbf{M}_A(\mathbf{k})|^2 \rangle} = R_A^{-1}(k, \omega = 0) \quad (5.7)$$

and, provided that k is sufficiently small for equation (5.3) to hold, the longitudinal and transverse ratios will be equal to a k independent constant. This expectation is borne out extremely well by the data (table 4). Thus, the vastly

Table 4. Relaxation times.

$k/k_{\min}$	$\Phi_T(k, \omega=0)/\text{ps}$	$\Phi_L(k, \omega=0)/\text{ps}$	$R_T(k)$	$R_L(k)$	f
1	3.2	0.16	7.2	8.8	0.6
2	3.7	0.14	8.0	7.7	0.4
3	4.3	0.14	8.0	7.0	0.4

different longitudinal and transverse relaxation times are shown to be a consequence of the long-range, static correlation which are contained in $\langle \mathbf{M}_A(\mathbf{k})^2 \rangle$ [1].

In order to describe the frequency dependence of ϵ , or equivalently of Φ_L and Φ_T , we need to prescribe the frequency dependence of $\tau_D(\omega)$ {equations (5.4)–(5.6)}. In dense fluids the molecular reorientation is slow compared with the relaxation of the angular velocity and intermolecular torques (the ‘diffusion limit’ [26]). At the low frequencies at which the dominant reorientational relaxation occurs, $\tau_D(\omega)$ should then be independent of frequency [1], i.e.

$$\tau_D(\omega) = \tau_D(0). \quad (5.8)$$

The range of validity of this simple diffusion approximation is shown in figure 4, which shows the result of using equation (5.8) in equation (5.4 *b*) with $\tau_D(0)$ assigned the value determined from the time integral of Φ_T (i.e. using equations (5.5) and (5.7)). The simple model adequately describes the shape of $\Phi_T(k, \omega)$ out to frequencies of about 20 cm^{-1} , the dominant relaxation in Φ_T (and $\epsilon(\omega)$) occurs over this range.

The central problem of dielectric relaxation theory is to relate the Debye relaxation time i.e. $\tau_D(\omega=0)$ to the rotational relaxation time of a single molecule. The normalized correlation function of a single dipole,

$$C_1(t) = \langle \mu_a(t) \mu_a \rangle / \langle \mu_a^2 \rangle, \quad (5.9)$$

rotating in a pure fluid may be written as [1]

$$C_1(\omega) = \{-i\omega + r(\omega)/(\mu^2/3)\}^{-1}, \quad (5.10 a)$$

where

$$r(\omega) = \int_0^\infty dt \exp[i\omega t] \langle \dot{\mu}_a(t_{(\mu_a)}) \dot{\mu}_a \rangle. \quad (5.10 b)$$

In equation (5.9) the time dependence is projected orthogonal to all the dipoles in the system. The relaxation time, τ_S , of $C_1(t)$, defined as the time integral, is

$$\tau_S = C_1(\omega=0) = (\mu^2/3)r(0)^{-1}. \quad (5.11)$$

The problem of relating $\tau_D(\omega=0)$ to the single particle relaxation time is thereby transcribed into that of relating the correlation functions which appear in equations (5.10 *b*) and (5.2); this is a useful transformation because R is a local quantity, independent of long range correlations.

The relationship of R to r is complicated by the fact that the projected time evolution operators in the two functions differ. However, it may be shown [1, 27, 28] when r is frequency independent that

$$R(\omega=0)=r(0)(1+f)N \quad (5.12)$$

where f measures the dynamic orientational correlation between molecules. The relationship between the Debye and single molecule relaxation times (cf. (5.5) and (4.1)) thus becomes

$$\tau_D/\tau_S=3\langle|\mathbf{M}_T|^2\rangle/(2N\mu^2(1+f)). \quad (5.13)$$

Relationships of this kind are commonly used in the interpretation of dynamical data from other fields of study; for example, to relate the Rayleigh scattering relaxation time to the reorientation time measured by N.M.R. or Raman scattering or to relate the widths of the quasielastic peaks in coherent and incoherent neutron scattering experiments. In these cases it is often found that the dynamic correlation parameter analogous to f may be neglected. In the dielectric case neglect of f leads to the 'Glarum-Powles' relationship between τ_D and τ_S [1].

The value of $(1+f)$ may be found by combining the data in tables 2 and 4, according to (5.13); the results are given in the last column of table 4. Even bearing in mind the likely inaccuracies in our relaxation times, it does appear that f is significantly different from zero. This behaviour has been noted in experimental [29] and simulation studies [21 (a)] of second rank orientational correlation functions.

5.3. High frequency behaviour

Although the diffusion model, in which the frequency dependence of τ_D is neglected, describes the behaviour of $\Phi_L(k, \omega)$ and $\Phi_T(k, \omega)$ up to fairly high frequencies (cf. figure 5), it cannot account for the shoulders and peaks which occur in these functions and which determine the shape of far infrared spectra. One question which should be resolved is whether the oscillation which occurs in the longitudinal correlation function should be identified as a 'dipolaron' oscillation, as predicted by Lobo *et al.* [8] and observed in a computer simulation of a Stockmayer fluid by Alder and Pollock [9]. Madden and Kivelson [1] have attempted to distinguish between the conditions under which dipolaron and librational oscillations may be observed. Dipolarons are expected to occur under conditions of rapid reorientation for materials with large permittivities; they are observed only in Φ_L , if librational oscillations occur they will be seen in Φ_L , Φ_T and C_1 at roughly the same frequency in each case.

Dipolarons are predicted to occur by theories which include the correct inertial short time behaviour in the correlation function. Within the Mori framework this may be accomplished by going to a 'two-variable theory' or to the 'second order continued fraction' {26, 28, 30}. The result of this extension of the simple diffusion result for Φ_L and Φ_T is

$$\Phi_A(k, \omega)=\left\{-i\omega+\frac{(\Omega_r^A(k))^2}{-i\omega+\Omega_l^2\Gamma^{-1}}\right\}^{-1} \quad (5.14)$$

where

$$(\Omega_r^A(k))^2 = \frac{\{2\}(2k_B T/I)N\mu^2}{\langle |\mathbf{M}_A(\mathbf{k})|^2 \rangle} \quad (5.15)$$

(and the factor $\{2\}$ is to be included only in the $A = T$ case). $\Omega_t^2 \Gamma^{-1}$ may be regarded as an angular velocity relaxation time [1, 26], by evaluating (5.14) for $A = T$, at $\omega = 0$ and comparing with (5.4 *b*) it may be related to τ_D , i.e.

$$\tau_D(\omega = 0) = \Omega_t^2 \Gamma^{-1} (\Omega_r^T)^{-2}. \quad (5.16)$$

Φ_A will exhibit inertial oscillations if equation (5.14) has poles at real frequencies ; this occurs if

$$\Omega_r^A > \Omega_t^2 \Gamma^{-1/2} \quad (5.17 a)$$

or, using (5.16), if

$$\Omega_r^A / \Omega_r^T > \tau_D \Omega_r^T / 2. \quad (5.17 b)$$

Values of Ω_r^T and Ω_r^L are given in table 5 ; since $\tau_D \sim 4.0$ ps this condition is not satisfied for either the longitudinal or transverse case. Consequently, the oscillation observed in Φ_L should not be identified as a dipolaron, it must be librational or torque driven.

Table 5. Dynamical parameters (all in radians/ps) used in figure 6.

$k/k_{\min}$	Ω_r^L	Ω_r^T	Ω_t	Γ	τ_D^{-1}
2	23.54	4.66	17.15	3.66	0.27

To generalize the Mori theory to describe librational oscillations it is necessary to go to a 'three variable theory' [26] in which the slowly relaxing character of the intermolecular torques, which are responsible for the libration, can be described. The result of this procedure is [1]

$$\Phi_A(\omega) = \left\{ -i\omega + \frac{(\Omega_r^A)^2}{-i\omega + \Omega_t^2(-i\omega + \Gamma)^{-1}} \right\}^{-1}. \quad (5.18)$$

Detailed expressions for Ω_t and Γ have been given elsewhere [1] ; broadly speaking, Ω_t^2 is related to the mean square torque and Γ is the torque relaxation rate. That these parameters are the same for $A = L$ or T follows from the same general considerations which lead to equation (5.3) [1]. Thus it is a prediction of the theory that both Φ_L and Φ_T can be described with the same parameter sets. The functional form of equation (5.18) has been widely used to fit far infrared absorption data [30] but the experimental lineshapes may be affected by induced dipoles. Here we shall test the ability of the equation to represent our (rigid dipole) data on Φ_L and Φ_T *simultaneously* ; so far as possible we shall constrain Ω_r^A , Ω_t and Γ to take values which can be independently obtained from the simulation.

Since the values of Ω_r^L and Ω_r^T are determined by equation (5.15) and since the ratio of Ω_t^2 to Γ is fixed by equation (5.16) the model for both Φ_L and Φ_T is completely specified if we choose a value for Ω_r . If the correlation between

the torques acting on different molecules is weak and if the mean squared torque acting is large (i.e. $\langle \mathbf{G}^2 \rangle^{1/2}/I \gg k_B T/I$, where $\mathbf{G}$ is the torque acting on a molecule) then [31, 21 (a)]

$$\Omega_l^2 = \langle \mathbf{G}^2 \rangle / 2Ik_B T. \quad (5.19)$$

We may readily determine this quantity, independently of the orientational correlation functions. The short time expansion of the normalized angular velocity correlation function, $C_\omega(t)$, reads [31]

$$C_\omega(t) = 1 - \frac{\langle \mathbf{G}^2 \rangle t^2}{4Ik_B T} + O(t^4) \quad (5.20)$$

so that the approximate value of Ω_l^2 may be obtained from the curvature of C_ω at $t=0$. The value of Ω_l obtained in this way is given in table 5 and the value for Γ obtained from equation (5.16) is also included for completeness.

The forms for $\Phi_L(\omega)$ and $\Phi_T(\omega)$ obtained from equation (5.18) using these parameters are shown in figure 6. Both the calculated functions show high frequency shoulders which are more prominent than are observed in the functions which emerge directly from the computer simulation (figure 5). The location of the shoulders is predicted quite accurately, reflecting the value of Ω_l and the accuracy of the prediction that Ω_l and Γ are 'A' independent. However the widths of these shoulders are considerably underestimated. This failure of the '3-variable theory' (i.e. equation (5.18)) has been noted in other computer simulation studies [21, 32] and discussed elsewhere [1]. A similar

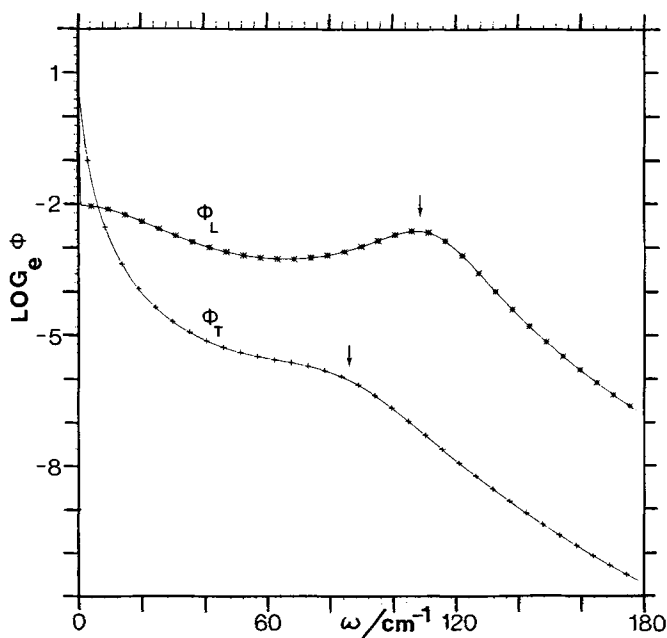


Figure 6. The longitudinal and transverse spectral densities predicted by the Mori theory (equation (5.18)), with the parameters given in table 5. The librational peaks are indicated by arrows; these are seen in the simulation spectra (figure 5), but are found to be much broader than predicted by the three-variable theory.

shortcoming is noted [33] when fitting experimental far I.R. data with parameters constrained so as to also describe the microwave relaxation. It may be attributed to the inadequacy of the simple exponential description of the torque relaxation implicit in the three variable theory. Thus the inability to offer a complete description of Φ_L and Φ_T at the frequencies of interest in far infrared absorption experiments does not lie in a failure to describe the effects of dipole-dipole interactions (i.e. 'dielectric effects'); that is, all 'dielectric effects' are explicitly described if we use equation (5.18) with Γ an 'A' independent but frequency dependent function. Rather, the fault lies in a more general lack of understanding of orientational motion at short times resulting from the difficulty of predicting the torque relaxation behaviour and consequent inability to predict the frequency dependence of $\Gamma(\omega)$.

6. SUMMARY

The work described in this and the succeeding paper was undertaken in order to aid the interpretation of the frequency dependence of the dielectric permittivity. To model realistically the dielectric properties of a real liquid we have performed the largest calculation which was possible with our resources. A six-site model of MeCN, shown in earlier work to accurately represent the real liquid, was simplified to a three-site model in such a way as to preserve the orientational structure of the fluid. The simulation contained 500 molecules and ran for 100 ps. The forces were truncated at one-half the box length and a large Ewald summation was used to treat the long range Coulomb interactions.

Although our major concern was to investigate dynamics we spent some time examining the static properties in order to show that the static properties (and thus the underlying dipole-dipole correlations) of the simulated fluids are consistent with those of a real fluid. Within sensible error limits we were able to show that the longitudinal and transverse components of the static permittivity are equal and (at most) slowly k dependent at the smallest accessible k values. Furthermore, the values for the permittivity calculated at finite k agree with that calculated at $k \equiv 0$. Values of the finite k 'Kirkwood factor', $g^S(k)$, were k independent over the range of small wavevectors studied, and agreed quite well with experimental values of the Kirkwood factor for acetonitrile [24]. The Kirkwood factor is much smaller than would be found in a Stockmayer fluid with the same reduced dipole moment, this was traced to the effect of the molecular shape on the short-range orientational structure of the fluid [3 (a)].

We have studied the time dependent correlation functions of the longitudinal (Φ_L) and transverse (Φ_T) components of the dipole density at finite k and of the total moment of the simulation cube (Φ_M). The shapes of Φ_M and Φ_T were the same, in accord with expectation for a simulation with Ewald summation, in the low k limit. Φ_L and Φ_T exhibited a dramatic difference in shape but the two functions could be related by the application of the theory of dielectric relaxation developed by Madden and Kivelson [1]. Although Φ_L exhibits a pronounced oscillatory feature which is only weakly present in Φ_T and the single particle function C_1 , this oscillation should not be attributed to a dipolaron oscillation of the type observed in simulations of the Stockmayer fluid [9].

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REFERENCES

- [1] MADDEN, P. A., and KIVELSON, D., 1984, *Adv. chem. Phys.*, **56** (in the press).
- [2] EDWARDS, D. M. F., and MADDEN, P. A., 1984, *Molec. Phys.*, **51**, 1163.
- [3] (a) BÖHM, H. J., McDONALD, I. R., and MADDEN, P. A., 1983, *Molec. Phys.*, **49**, 347.
(b) BÖHM, H. J., LYNDEN-BELL, R. M., MADDEN, P. A., and McDONALD, I. R., 1984, *Molec. Phys.*, **51**, 761.
- [4] STELL, G. S., PATEY, G., and HØYE, J. S., 1981, *Adv. chem. Phys.*, **48**, 183.
- [5] ALDER, B. J., and POLLOCK, E. L., 1981, *A. Rev. phys. Chem.*, **32**, 311.
- [6] FRENKEL, D., 1980, *Intermolecular Spectroscopy and the Dynamical Properties of Dense Systems*, edited by J. Van Kranendonk (North-Holland).
- [7] POLLOCK, E. L., and ALDER, B. J., 1980, *Physica A*, **102**, 1.
- [8] LOBO, R., ROBINSON, J. E., and RODRIGUEZ, S., 1973, *J. chem. Phys.*, **59**, 5992.
- [9] POLLOCK, E. L., and ALDER, B. J., 1981, *Phys. Rev. Lett.*, **46**, 950.
- [10] (a) DE LEEUW, S. W., PERRAM, J. W., and SMITH, E. R., 1980, *Proc. R. Soc. A*, **373**, 27.
(b) FELDERHOF, B. U., 1980, *Physica A*, **101**, 275.
- [11] IMPEY, R. W., MADDEN, P. A., and TILDESLEY, D. J., 1981, *Molec. Phys.*, **44**, 1319.
- [12] TITULAER, U. M., and DEUTCH, J. M., 1974, *J. chem. Phys.*, **60**, 1502.
- [13] COX, S. R., and WILLIAMS, D. E., 1981, *J. comput. Chem.*, **2**, 304.
- [14] MAITLAND, G. C., RIGBY, M., SMITH, E. B., and WAKEHAM, W. A., 1981, *Intermolecular Forces* (Clarendon Press).
- [15] SINGER, K., TAYLOR, A., and SINGER, J. V. L., 1977, *Molec. Phys.*, **33**, 1757.
- [16] KLEIN, M. L., McDONALD, I. R., and O'SHEA, S. F., 1978, *J. chem. Phys.*, **69**, 63.
- [17] EDWARDS, D. M. F., 1983, Ph.D. Thesis (University of Cambridge).
- [18] BERTAGNOLLI, H., and ZEIDLER, M. D., 1978, *Molec. Phys.*, **35**, 177.
- [19] STONE, A. J., 1978, *Molec. Phys.*, **36**, 241. BLUM, L., and TORRUELLA, A. J., 1972, *J. chem. Phys.*, **56**, 303.
- [20] HURLE, R. L., and WOOLF, L. A., 1982, *J. chem. Soc. Faraday Trans. I*, **78**, 2233.
- [21] (a) TILDESLEY, D. J., and MADDEN, P. A., 1982, *Molec. Phys.*, **48**, 129. (b) IMPEY, R. W., MADDEN, P. A., and McDONALD, I. R., 1982, *Molec. Phys.*, **46**, 513.
- [22] GRIFFITHS, J., 1977, *Vibrational Spectra and Structure*, edited by J. Durig (Elsevier), p. 273.
- [23] STEINHAUSER, O., 1982, *Molec. Phys.*, **46**, 827.
- [24] BUCKINGHAM, A. D., 1967, *Faraday Discuss. chem. Soc.*, **43**, 205 ; 1956, *Proc. R. Soc. A*, **238**, 235.
- [25] MORI, H., 1965, *Prog. theor. Phys.*, **33**, 423 and **34**, 399.
- [26] KIVELSON, D., and KEYES, T. F., 1972, *J. chem. Phys.*, **57**, 4599.
- [27] KEYES, T., and KIVELSON, D., 1971, *J. chem. Phys.*, **56**, 1057.
- [28] KIVELSON, D., and MADDEN, P. A., 1975, *Molec. Phys.*, **30**, 1749.
- [29] MADDEN, P. A., BATTAGLIA, M. R., COX, T. I., PIERENS, R. K., and CHAMPION, J., 1980, *Chem. Phys. Lett.*, **76**, 604.
- [30] EVANS, M., EVANS, G., and DAVIES, R., 1980, *Adv. chem. Phys.*, **44**, 255.
- [31] ST. PIERRE, A. G., and STEELE, W. A., 1981, *Molec. Phys.*, **43**, 123.
- [32] LYNDEN-BELL, R. M., and McDONALD, I. R., 1981, *Molec. Phys.*, **43**, 1429.
- [33] YARWOOD, J., and EVANS, M. W. (personal communication).